1 Basic theorems

This section develops the theory of M_A in the setting of **??**: its structure as the law of a random theory and its updating rules (Section 1.1), its computability (Section 1.2), dominance and invariance (Section 1.3), loss bounds (Section 1.4), the failure of a coding theorem (Section 1.5), the reasons to prefer proof-requiring compatibility (Section 1.6), and what the posterior does in the limit (Section 1.7). Proofs not given in place are in **??**.

1.1 Structure and updating

M_A is the law of a random consistent theory, read through what it proves. The first result makes this precise; the companion paper's notions of a weak semimeasure (its Definition 3.2: the masses of the extensions of D by one more datum sum to at most the mass of D) and of a containment functional (its Definition 3.4: $D \mapsto \Pr[F \supseteq D]$ for a random partial function F) are the reference points.

Proposition 1.1 (Structure)**.** *Let G be a theory machine.*

(a) If $\nu_G(D) > 0$, then D is consistent. $M_A(D) > 0$ iff D is consistent; $M_A^3(D) > 0$ iff D is consistent and $B \cup D^{\pm}$ decides no ι-labelled sentence.

(b) For two-valued D and $\varphi \notin \operatorname{dom} D$: $\nu_G(D+(\varphi,1))+\nu_G(D+(\varphi,0))+\nu_G^3(D+(\varphi,\iota)) = \nu_G(D)$. So ν_G is a weak semimeasure on data sets, and for M_A the defect at φ divided by $M_A(D)$ is $\operatorname{Indep}_D(\varphi)$.

(c) $\nu_G(D)$ depends on D only through $\operatorname{Cn}(D^{\pm})$. If $B \cup D^{\pm} \vdash \varphi^b$ and $\varphi \notin \operatorname{dom} D$, then $\nu_G(D + (\varphi, b)) = \nu_G(D)$ and $\nu_G(D + (\varphi, 1-b)) = 0$: entailed data cost nothing, contradicting data are fatal.

(d) For every consistent reveal sequence, in any order and with repeats, $\sum_{t\le n} -\log M_A(y_t \mid D_{t-1}, \varphi_t) = L_A(D_n) - L_A(\emptyset)$, which depends only on $\operatorname{Cn}(D_n^{\pm})$.

(e) $0 < M_A(\emptyset) = 1 - \operatorname{Inc}_0 < 1$.

(f) For three-valued D and $\varphi \notin \operatorname{dom} D$: $\sum_{y\in Y_3} \nu_G^3(D + (\varphi, y)) = \nu_G^3(D)$, and $\nu_G^3(\emptyset) = \nu_G(\emptyset)$. So the conditionals of M_A^3 are probability distributions on Y_3.

(g) $\nu_G^3(D) = \mathbb{U}\{\omega \in \operatorname{Cons}^G : \operatorname{tri}_\omega^G \supseteq D\}$, and $\nu_G(D) = \mathbb{U}\{\omega \in \operatorname{Cons}^G : \operatorname{tri}_\omega^G \supseteq D\}$ for two-valued D. So, when $\nu_G(\emptyset) > 0$, $\nu_G^3/\nu_G(\emptyset)$ is the containment functional of the random total function $\operatorname{tri}_\omega^G$ under $\mathbb{U}(\cdot \mid \operatorname{Cons}^G)$, and $\nu_G/\nu_G(\emptyset)$ is that of the random partial function obtained by reading ι as "undefined".

On the event E_D^G the random theory is consistent, so at φ exactly one of three things happens: it proves φ, it refutes φ, or it decides neither; these are the three events of (b). The other parts follow from the fact that O_χ^G depends only on the B-equivalence class of χ, and (e) from the tapes that begin with programs for the empty theory and for $\{\bot\}$ (**??**). By (g), M_A is a (sub-probability) containment functional, a strictly stronger property than being weak (the companion paper's Example 3.5, and Proposition 1.21(e) below). Its losses are those of a Bayes mixture over random theories, and its predictions are posterior averages of what individual theories prove.

Proposition 1.2 (Proof-requiring updating is geometric conditioning)**.** *(a) For every D with $M_A(D) > 0$ and every χ: $\operatorname{Bel}_D(\chi) = \operatorname{Bel}_0(\psi_D \wedge \chi)/\operatorname{Bel}_0(\psi_D) = \widehat{\operatorname{Bel}}_0(\psi_D \wedge \chi)/\widehat{\operatorname{Bel}}_0(\psi_D)$.*

(b) Bel_0 *and every* Bel_D *are totally monotone on the Lindenbaum algebra of* B*: for all* $\chi_1, \dots, \chi_k$, $\mathrm{Bel}_D(\bigvee_{i\le k}\chi_i) \ge \sum_{\emptyset\neq I\subseteq[k]}(-1)^{|I|+1}\mathrm{Bel}_D(\bigwedge_{i\in I}\chi_i)$.

(c) There is a probability P_D *on the complete consistent extensions of* B *(a Borel probability on the Stone space) with* $\mathrm{Bel}_D(\chi) \le P_D(\chi) \le \mathrm{Pl}_D(\chi)$ *for every* χ.

So Bel_D is a belief function in the sense of Shafer (1976), with random focal set $\mathrm{Mod}(B \cup A_\omega)$ and plausibility Pl_D; Bel_0 is the probability of provability, as Pearl puts it (Pearl, 1988). Part (a) is the *geometric rule* of conditioning of $\widehat{\mathrm{Bel}}_0$, usually attributed to Suppes and Zanotti (Suppes and Zanotti, 1977): keep the focal sets inside the conditioning event and renormalise. It holds because a theory proves ψ_D and χ iff it proves $\psi_D \wedge \chi$; (b) is inclusion–exclusion for the events O_{χ_i}, whose intersections are again of this form; (c) completes each consistent random theory by a fixed Lindenbaum construction. Parts (b) and (c) are the one-tape forms of Propositions 2.18 and 2.19 of Hänni and Claude (Anthropic) (2026), which are stated for countable classes of theories.

Proposition 1.3 (Consistency-requiring updating is Dempster conditioning)**.** *Let* $\psi := \psi_D$ *and* $M_A^{\mathrm{nc}}(D) > 0$*. Then* $M_A^{\mathrm{nc}}(D) = \mathrm{Pl}_0(\psi)$ *and*

$$\mathrm{Bel}_D^{\mathrm{nc}}(\varphi) = \frac{\mathrm{Bel}_0(\varphi \vee \neg\psi) - \mathrm{Bel}_0(\neg\psi)}{\mathrm{Bel}_0(\top) - \mathrm{Bel}_0(\neg\psi)} = \frac{\widehat{\mathrm{Bel}}_0(\varphi \vee \neg\psi) - \widehat{\mathrm{Bel}}_0(\neg\psi)}{1 - \widehat{\mathrm{Bel}}_0(\neg\psi)}.$$

The right-hand side is Dempster's rule of conditioning (Dempster, 1967; Shafer, 1976). The proof is the deduction theorem: for consistent $B \cup A$, $B \cup A \cup \{\psi\}$ is consistent iff $B \cup A \nvdash \neg\psi$, and proves φ iff $B \cup A \vdash \varphi \vee \neg\psi$. In the random-set picture with focal sets $F_\omega := \mathrm{Mod}(B \cup A_\omega)$, proof-requiring updating keeps the focal sets contained in $\mathrm{Mod}(\psi)$, while Dempster's rule replaces each F_ω by $F_\omega \cap \mathrm{Mod}(\psi)$ and drops the empty ones; both renormalise. Under proof-requiring updating a datum must be explained by the hypothesis; under consistency-requiring updating it is added to it.

1.2 Computability

Proposition 1.4. $M_A(D) = \mathbb{U}(O_{\psi_D}) - \mathrm{Inc}_0$*, where* $\mathbb{U}(O_{\psi_D})$ *is left-c.e. uniformly in* D *and* $\mathrm{Inc}_0 = \mathbb{U}(O_\perp)$ *is left-c.e.*

Proof. This is **??**(a), (c) for $G = G_U$. □

The lsc function $D \mapsto \mathbb{U}(O_{\psi_D})$ is not a semimeasure, since it gives an inconsistent theory credit for every datum. The correction that turns it into M_A is one real number, and that real is as complex as a left-c.e. real can be.

Theorem 1.5 (The inconsistency probability is an Ω-like real)**.** *(a)* Inc_0 *is not computable. Hence* $M_A(\emptyset) = 1 - \mathrm{Inc}_0$ *is right-c.e. and not left-c.e., and* $D \mapsto M_A(D)$ *is not lower semicomputable.*

(b) For every left-c.e. $\alpha \in [0,1]$ *there are a* V*-program* p_α *and a left-c.e. real* γ_α *with* $\mathrm{Inc}_0 = 2^{-|p_\alpha|}\alpha + \gamma_\alpha$*. Consequently* $\alpha \le_S \mathrm{Inc}_0$ *for every left-c.e.* α*:* Inc_0 *is Solovay complete.*

(c) Inc_0 *is Martin-Löf random.*

For a left-c.e. α, a theory machine emits $\perp$ exactly on the tapes ω whose binary real $0.\omega$ lies below α (up to a null set), so its inconsistency probability is α; splitting the tapes by their first program gives (b). An α whose binary expansion is the halting set gives (a), and $\alpha = \Omega_V$ with

Demuth's theorem that randomness passes to $2^{-k}\alpha+\gamma$ (**??**) gives (c), since Ω_V is random (Chaitin, 1975). The argument is the standard one that Ω is Solovay complete (Solovay reducibility $\leq_S$ as in Solovay, 1975; Downey and Hirschfeldt, 2010), applied to the event that the random theory is inconsistent; reals of the form "the probability that a universal machine has a given behaviour" are studied in Becher et al. (2006) and Barmpalias et al. (2016). By the characterisation of the random left-c.e. reals (Kučera and Slaman, 2001; Calude et al., 2001), Inc_0 is an Ω-number; we do not use this.

Every machine law of SOI is lsc, and M_F is lsc with $\emptyset'$-computable conditionals (Propositions 3.7 and A.10 of the companion paper). M_A keeps the second property and loses the first, by exactly one real: its values and conditionals are $\emptyset'$-computable, and $M_A+\mathrm{Inc}_0$ is lsc (**??**). Function machines are consistent by construction, while the consistency of a theory machine's output is a Π^0_1 property of the tape. When consistency of finite sets of sentences with B is not r.e., none of the versions is lsc (**??**).

1.3 Dominance and invariance

Theorem 1.6 (Mixture identity and dominance)**.** *For every two-valued D,*

$$M_A(D) = \sum_{p\in\operatorname{dom} V} 2^{-|p|}\nu_{G_{V(p)}}(D) + (1-\Omega_V)\mathbf{1}[B\vdash\psi_D] = \sum_i \mathbf{m}(i)\,\nu_{G_i}(D) + (1-\Omega_V)\mathbf{1}[B\vdash\psi_D],$$

and the same holds for M^3_A, the laws $\nu^3_{G_i}$ and the indicator that B proves ψ_D and decides no ι-labelled sentence. Hence $M_A \geq 2^{-K(\nu)}\nu$ for every theory-machine law ν, and $M^3_A \geq 2^{-K(G)}\nu^3_G$ for every theory machine G. In particular:

- *(i) $M_A(D) \geq 2^{-K(e)-c}\mathbf{1}[B\cup\mathsf{W}_e$ consistent, $\vdash\psi_D]$ for every index e; consequently, for every r.e. T with $B\cup T$ consistent, $\mathbb{U}_A[T] \geq 2^{-c}\mathbf{m}_{\equiv}(T) \geq 2^{-K_{\equiv}(T)-c}$ and $\mathbb{U}_A(\mathrm{Ext}T) \geq 2^{-c}\mathbf{m}(\mathrm{Ext}T)$;*
- *(ii) $M_A(D) \geq 2^{-K(f)-c}\mathbf{1}[f\supseteq D]$ for every partial computable labeller $f:\mathrm{Sent}\rightharpoonup\{0,1\}$ with $B\cup\Gamma_f$ consistent, where $\Gamma_f := \{\varphi^{f(\varphi)} : \varphi\in\operatorname{dom} f\}$;*
- *(iii) for every lsc semimeasure v on indices, $M_A(D) \geq 2^{-K(v)-c}\sum_e v(e)\mathbf{1}[B\cup\mathsf{W}_e$ consistent, $\vdash\psi_D]$; in particular $M_A \geq 2^{-c}W_A$;*
- *(iv) for an r.e. T and a computable family (α_j) of sentences, the law of T plus the fair-coin facts $\alpha_j^{\omega_j}$ (Example (iii) after **??**) is dominated with constant $2^{-K(T,(\alpha_j))-c}$.*

The cylinders $[p]$, $p\in\operatorname{dom} V$, are disjoint, and on $[p]$ the universal machine emits what $G_{V(p)}$ emits on the rest of the tape; so $\mathbb{U}(E_D\cap[p]) = 2^{-|p|}\nu_{G_{V(p)}}(D)$, and the tapes with no program emit the empty theory. The rest uses machines with indices computable from e, f, v or $(T,(\alpha_j))$ (**??**). The mixture identity and the dominance argument are the standard ones for universal mixtures (Zvonkin and Levin, 1970; Theorem 3.8 of the companion paper); what is new is the object they are applied to. A consequence is that $L_A(D) = \min_G[K(G) - \log\nu_G(D)] \pm c$, the minimum over theory machines: SAI is the best two-part code over theory machines, randomised theories included (**??**).

Proposition 1.7 (Choice of the universal machine)**.** *Call a theory machine G'* universal *if for every theory machine G there is a string p_G with $A^{G'}_{p_G\omega} = A^G_\omega$ for all ω. If G' is universal, then $2^{-c}M_A \leq \nu_{G'} \leq 2^cM_A$, and the same for the three-valued laws, with c depending only on G_U and G'.*

Proof. $M_A \geq 2^{-K(G')}\nu_{G'}$ by Theorem 1.6. Conversely, the tapes $p_{G_U}\omega'$ give G' the axiom set $A_{\omega'}$, so $\nu_{G'}(D) \geq 2^{-|p_{G_U}|}M_A(D)$; the three-valued case is the same. □

So posteriors change by at most a constant factor, and semimeasure-convention losses by an additive constant, when the universal machine is changed. The background is a prior shift of the same kind:

Proposition 1.8 (The background is a prior shift)**.** *Let B, B' be decidable and consistent with $B \subseteq \mathrm{Cn}_\emptyset(B')$ (plain logical consequence), and write M_A^B, $M_A^{B'}$ for SAI with these backgrounds. Then $M_A^B(D) \geq 2^{-K(B')-c}M_A^{B'}(D)$ for every D, with $K(B')$ the complexity of an index of B'; likewise $W_A^B \geq 2^{-K(B')-c}W_A^{B'}$ and $K_A^B \leq K_A^{B'} + K(B') + c$. The converse fails: a D consistent with B and inconsistent with B' has $M_A^{B'}(D) = 0 < M_A^B(D)$.*

Proof. The machine $\{(\epsilon, \chi) : \chi \in B'\} \cup G_U$ emits $B' \cup A_\omega$ on ω, and under background B the set $B \cup B' \cup A_\omega$ is equivalent to $B' \cup A_\omega$; so its law under background B is $M_A^{B'}$ (G_U does not depend on the background), and Theorem 1.6 applies. The same map on single programs gives the other two statements. □

So building knowledge into the background gains at most $K(B') + c$ bits on any data. A change of language costs a constant too, provided the translation preserves what is proved:

Theorem 1.9 (Faithful translations)**.** *Let $(\mathcal{L}, B)$ and $(\mathcal{L}', B')$ be two settings and $\tau : \mathrm{Sent}_\mathcal{L} \to \mathrm{Sent}_{\mathcal{L}'}$ computable with*

(T1) for all $\Gamma \subseteq \mathrm{Sent}_\mathcal{L}$ and φ: $B \cup \Gamma \vdash \varphi$ iff $B' \cup \tau[\Gamma] \vdash \tau(\varphi)$;

(T2) $B' \vdash \tau(\neg\varphi) \leftrightarrow \neg\tau(\varphi)$ for all φ.

For a data set D on $\mathcal{L}$ let $M'_A(\tau D) := \mathbb{U}\{\omega : B' \cup A'_\omega$ consistent and $\vdash \tau(\varphi)^{D(\varphi)}$ for all $\varphi \in \mathrm{dom}\, D\}$, with A' the universal theory machine of $\mathcal{L}'$, and define $W'_A(\tau D)$, $K'_A(\tau D)$ likewise. Then for every D

$$2^{-K(\tau)-c}M_A(D) \leq M'_A(\tau D) \leq 2^{K(\tau)+c}M_A(D),$$

the same holds for W_A, and $|K'_A(\tau D) - K_A(D)| \leq K(\tau) + c$. Here c depends on both settings.

One direction runs G_U and emits $\tau[A_\omega]$; by (T1) and (T2) its event for τD is exactly E_D. The other pulls a theory A' back to $\{\varphi : B' \cup A' \vdash \tau(\varphi)\}$, which is consistent when A' is and proves what A' proves (**??**).

Corollary 1.10 (Adding vocabulary)**.** *Let $\mathcal{L} \subseteq \mathcal{L}'$, keep the background B, and let τ be the inclusion. Then (T1) and (T2) hold, and for data in the old vocabulary $M_A^{\mathcal{L}'}(D)$ and $M_A^\mathcal{L}(D)$ agree up to a factor $2^{\pm c}$, W_A likewise, $K_A^{\mathcal{L}'}(D) = K_A^\mathcal{L}(D) \pm c$, and predictive probabilities of $\mathcal{L}$-sentences agree up to $2^{\pm 2c}$.*

(T1) holds because every $\mathcal{L}$-structure expands to an $\mathcal{L}'$-structure. In the unbounded universal setting, then, new vocabulary buys at most a constant: the pull-back of a theory is the set of its consequences in the old vocabulary, an r.e. set, which the universal machine enumerates. This is the effective content of Craig's theorem (Craig, 1953) inside the prior. **??** extends it to class-level posterior odds, to penalised versions and to an exact tape-by-tape identity; where vocabulary does pay (finite theories, charged deduction) is the subject of **??**.

1.4 Loss bounds

Theorem 1.11 (Loss against a theory that proves the data)**.** *Let $(\varphi_t, b_t)_{t\le n}$ be any consistent reveal sequence, chosen in any way (in advance, adaptively, adversarially or non-computably; in any order; with repeats), and let T be an r.e. theory with $B \cup T$ consistent and $B \cup T \vdash \varphi_t^{b_t}$ for all $t \le n$. Then in the semimeasure convention*

$$L_A(D_n) - L_A(\emptyset) \;\le\; L_A(D_n) \;\le\; -\log \mathbb{U}_A(\mathrm{Ext}T) \;\le\; -\log \mathbf{m}(\mathrm{Ext}T) + c \;\le\; -\log \mathbf{m}_{\equiv}(T) + c \;\le\; K_{\equiv}(T) + c,$$

and also $L_A(D_n) \le -\log W_A(D_n) + c \le K_A(D_n) + c$.

Proof. Every consistent extension of T proves every datum, so E_{D_n} contains the event that $B \cup A_\omega$ is consistent and $\mathrm{Cn}(A_\omega) \supseteq \mathrm{Cn}(T)$. Then use Theorem 1.6(i), $[T] \subseteq \mathrm{Ext}T$ (as $B \cup T$ is consistent), $\mathbf{m}_{\equiv}(T) \ge 2^{-K_{\equiv}(T)}$ and $L_A(\emptyset) \ge 0$; the left side is the cumulative loss by Proposition 1.1(d). The last chain is Theorem 1.6(iii) and $W_A \ge 2^{-K_A}$. □

The complexity that appears is not that of an axiom set and not a property of the data's syntax: it is the algorithmic probability of the class of all consistent extensions of T, bounded by $K_{\equiv}(T)$, the complexity of T up to deductive equivalence. So the bound favours the weakest theory that decides the data ($\mathrm{Cn}T \subseteq \mathrm{Cn}T'$ implies $\mathrm{Ext}T' \subseteq \mathrm{Ext}T$); a theory that is not finitely axiomatisable, such as PA, has small $K_{\equiv}$ although every axiomatisation of it is infinite; and of two equivalent axiomatisations the simpler one counts. Compared with the bound $K(f) + c$ of the companion paper (its Theorem 5.1(b)) for data from a partial computable f, a program is replaced by a theory up to equivalence, and the theory need not be complete, only decide the sentences asked.

Proposition 1.12 (Laws plus exceptions)**.** *(a) For every theory-machine law ν: $L_A(D_n) \le -\log \nu(D_n) + K(\nu)$.*

(b) For every r.e. T and every sub-data-set $E \subseteq D_n$ with $B \cup T \cup E^{\pm}$ consistent and $B \cup T \cup E^{\pm} \vdash \psi_{D_n}$: $L_A(D_n) \le K_{\equiv}(T) + K(E) + c$.

(c) Let $(\alpha_j)_{j\ge 1}$ be a computable family of sentences and T r.e. such that $B \cup T \cup \{\alpha_j^{x_j} : j \ge 1\}$ is consistent for every $x \in \{0,1\}^{\mathbb{N}}$. If every datum of D_n is proved by T or is one of the α_j with some label, and m distinct j occur, then $L_A(D_n) \le K(T, (\alpha_j)) + m + c$.

Proof. (a) is Theorem 1.6. (b) For $e \in [T]$, the set $\mathsf{W}_e \cup \{\psi_E\}$ has an index computable from (e, E), is consistent with B and proves ψ_{D_n}. (c) The machine that emits T and $\alpha_j^{\omega_j}$ for all j always emits a consistent theory, which proves D_n when ω agrees with the m labels of the α_j-data, an event of probability 2^{-m}; apply (a). □

Taking for E in (b) the data that T does not prove gives a two-part bound, a law plus listed exceptions, in which one pays $K(E)$, including the positions of the exceptions. In (c) each exception costs one bit, without paying for which atoms were exceptional. **??** extends (c) from atoms a law is silent about to answers it only constrains.

Proposition 1.13 (Mistake bounds)**.** *(a) Under the hypotheses of Theorem 1.11, the predictor that outputs the most probable of $1, 0, \iota$ under $(\mathrm{Bel}, \mathrm{Dis}, \mathrm{Indep})_{D_{t-1}}(\varphi_t)$ (ties broken arbitrarily; the stored label at a repeat) makes at most $-\log(\mathbb{U}_A(\mathrm{Ext}T)/M_A(\emptyset)) \le K_{\equiv}(T) + c$ mistakes, a mistake being any output other than b_t (ι included).*

(b) *Fix a well-order of* $\operatorname{dom} V$ *that refines length. The* Kolmogorov predictor *takes at step* t *the first program* p_t *with* $B \cup \mathsf{W}_{V(p_t)}$ *consistent and proving* $\psi_{D_{t-1}}$*, predicts* 1 *(resp.* 0*) if* $\mathsf{W}_{V(p_t)}$ *proves* φ_t *(resp.* $\neg\varphi_t$*), and abstains otherwise. If* $k := \sup_n K_A(D_n) < \infty$ *(for instance* $k \le K_{\equiv}(T) + c$ *under the hypotheses of Theorem 1.11), it errs or abstains at most* 2^k *times.*

(c) *The order hypothesis in (b) is needed: for every* m *there are a well-order of* $\operatorname{dom} V$ *of type* ω *and a data set* D *with* $K_A(D) \le K(m) + c \le 2\log m + c'$ *on which the Kolmogorov predictor with that order errs or abstains* $2^m - 1$ *times.*

In (a) each mistake costs at least one bit of cumulative loss (the halving argument, Corollary 5.4 of the companion paper). In (b) a program that errs or abstains is incompatible with all later data and is never chosen again, and Kraft's inequality allows at most 2^k programs of length at most k. Part (c) refutes the natural strengthening of (b) to every well-order: an order that lists decoys first defeats it, by the construction of Theorem 1.27 (**??**).

With three-valued data the questions may be chosen by any rule, and the predictions merge with those of the true theory machine:

Theorem 1.14 (Merging for three-valued data)**.** *Let* G *be a theory machine with* $\nu_G(\emptyset) = 1$*,* $\omega \sim \mathbb{U}$ *and* $A := A^G_\omega$*. Let* (u_t) *be random variables independent of* ω *(with any joint law), and let the questions be* $\varphi_t := s_t(h_{t-1}, u_t)$ *for arbitrary measurable rules* s_t*, where* $h_{t-1} = (\varphi_1, y_1, \dots, \varphi_{t-1}, y_{t-1})$ *and* $y_t \in Y_3$ *is the trichotomy label of* φ_t *under* A*. Let* $D_t := \{(\varphi_s, y_s) : s \le t\}$ *and let* $d_t := \mathrm{KL}\left(\nu^3_G(\cdot \mid D_{t-1}, \varphi_t) \,\middle\|\, M^3_A(\cdot \mid D_{t-1}, \varphi_t)\right)$ *in bits (*$d_t := 0$ *at a repeat). Then for every* n

$$\sum_{t \le n} \mathbb{E} d_t = \mathbb{E} \log \frac{\nu^3_G(D_n)}{M^3_A(D_n)} + \log M_A(\emptyset) \;\le\; K(G) + \log M_A(\emptyset) \;<\; K(G).$$

Hence $d_t \to 0$ *almost surely, and* $\sum_t \mathbb{E}\|\nu^3_G(\cdot \mid D_{t-1}, \varphi_t) - M^3_A(\cdot \mid D_{t-1}, \varphi_t)\|_1^2 \le 2\ln 2 \cdot K(G)$*.*

Given the history and the side randomness, the next label has law $\nu^3_G(\cdot \mid D_{t-1}, \varphi_t)$, because the history constrains the tape only through the event $\mathrm{tri}^G_\omega \supseteq D_{t-1}$; the log-ratio then telescopes, and dominance bounds it (**??**). With two-valued data the question rule would have to see A (ask only questions A decides) for the two-valued conditionals to be the data law, and the expected bound can fail, as in the companion paper's remark after its Theorem 5.3; the pathwise bound Proposition 1.12(a) still holds. The three-valued analogue of Theorem 1.11 replaces the extensions of T by its deductive class:

Corollary 1.15 (Three-valued loss)**.** *If* D_n *is three-valued and labelled by the trichotomy of an r.e.* T *with* $B \cup T$ *consistent, then in any reveal order the cumulative loss in the three-valued convention is* $-\log M^3_A(D_n) + \log M_A(\emptyset) \le -\log \mathbb{U}_A[T] \le K_{\equiv}(T) + c$*.*

Three-valued data penalise theories that decide more than T. Lower bounds come from coding:

Proposition 1.16 (Lower bound)**.** *For every data set* D*,* $L_A(D) \ge K(b_D \mid \operatorname{dom} D) - c$*, where* b_D *is the vector of two-valued labels in canonical order. So on fair-coin labels of any fixed* n *sentences,* $\mathbb{E}L_A \ge n - c$ *in the semimeasure convention, whatever the sentences.*

The first claim is Proposition 1.17(b) below; the second follows by Gibbs' inequality, since $\sum_b 2^{-K(b|\operatorname{dom} D)} \le 1$. With logically dependent sentences, inconsistent labellings have $L_A = \infty$, and the bound holds trivially for them.

1.5 No coding theorem

For SOI, the one-tape, coin-free and Kolmogorov versions differ by more than a constant (the companion paper's Theorem 4.9). The same holds for axiom systems. The bounds that do hold form a chain.

Proposition 1.17 (Coding lower bound and the chain)**.** *(a) Let S be a finite set of pairwise B-incompatible sentences and $\sigma \in S$. Then $-\log \mathrm{Bel}_0(\sigma) = -\log M_A(\{(\sigma,1)\}) \geq K(\sigma \mid S) - c$.*

(b) For every data set D, $L_A(D) \geq K(b_D \mid \mathrm{dom}\, D) - c$.

(c) For every consistent D and every finite set S of pairwise B-incompatible sentences with $\psi_D \in S$,

$$2^{-K_A(D)} \leq W_A(D) \leq 2^c M_A(D) \leq 2^{K(S)+c}\, 2^{-K_A(D)}.$$

With S the $2^{|\mathrm{dom}\, D|}$ label conjunctions on the two-valued part of $\mathrm{dom}\, D$, $K(S) \leq K(\mathrm{dom}\, D) + c$.

M_A is not lsc (Theorem 1.5), so the coding theorem is applied in (a) to an auxiliary lsc semimeasure on S: the probability that a dovetailed proof search from the emitted axioms finds a given member of S first. A consistent theory proves at most one member of S, so this semimeasure is at least Bel_0 on S (**??**). The slack $K(S)$ is the complexity of a set of pairwise incompatible sentences containing the data's conjunction, for instance the direct answers of a question (though S need not be exhaustive), in place of the companion paper's $K(\mathrm{dom}\, D)$ (its Theorem 4.4). For $S = \{\psi_D, \neg\psi_D\}$ the slack is $K(\psi_D)$, which is useless; for "which of these m incompatible laws holds" it is $K(m) + O(1)$ when S is computable from m.

Theorem 1.18 (No coding theorem for axiom systems)**.** *In the literal setting, for every $d \geq 2$ there are distinct $x_1, x_2 \in \{0,1\}^d$ such that $D := \{(R(t_{x_1}), 0), (R(t_{x_2}), 1)\}$ satisfies*

(i) $K_A(D) \geq \ell_d > \log d - 2$, with $\ell_d := \lceil \log(d/\log 3) \rceil - 1$, and $W_A(D) \leq \Omega_V^{\geq \ell_d}$;

(ii) $-\log W_A(D) \leq 2 + K(\lceil \log(d+1) \rceil) + c$;

(iii) $-\log M_A(D) \leq 2 + c$.

Parts (ii) and (iii) hold for every data set of two literals on distinct words of length at most d. Hence $K_A(D) + \log W_A(D) \geq \log d - K(\lceil \log(d+1) \rceil) - c - 4 \to \infty$ and $M_A(D)/W_A(D) \geq 2^{-c-2}/\Omega_V^{\geq \ell_d} \to \infty$: no two of 2^{-K_A}, W_A, M_A agree up to a constant factor.

Colour each word of length d by what the fewer than 2^{ℓ_d} programs of length below ℓ_d prove about its literal; there are fewer colours than words, so two words share a colour, and no short program separates them. A coin per atom, on the other hand, proves a given labelling of two atoms with probability $\frac{1}{4}$ (**??**). This is the companion paper's Theorem 4.9 for axiom systems, proved directly with theory machines, and the mechanism is the same: deterministic programs (here r.e. axiom sets) cannot separate atoms cheaply, while a coin can. The data set is the single sentence ψ_D, so the natural strengthening that a coding theorem holds for single sentences fails. On literal data the two-point rate of the companion paper transfers (**??**). For whole theories, $K_{\equiv}$ and $-\log \mathbf{m}_{\equiv}$ agree up to a factor 3 and logarithmic terms (**??**).

1.6 Why proof-requiring compatibility is primary

Proof-requiring compatibility is the stronger requirement, and it has the learning guarantees: dominance (Theorem 1.6), the loss bound $K_{\equiv}(T) + c$ (Theorem 1.11), mistake bounds and merging

(Proposition 1.13 and Theorem 1.14), and the structure of a law of random theories (Proposition 1.1). The weaker requirements fail as follows. Let $H_\omega := B \cup A_\omega \cup D^{\pm}$, the hypothesis together with the data.

Proposition 1.19 (The 0/1 consistency-requiring weight does not discriminate)**.** *(a) For $\varphi \notin \operatorname{dom} D$: $M^{\mathrm{nc}}_A(D{+}(\varphi,1)){+}M^{\mathrm{nc}}_A(D{+}(\varphi,0)) = M^{\mathrm{nc}}_A(D){+}\mathbb{U}\{\omega : H_\omega$ consistent and decides neither φ nor $\neg\varphi\} \geq M^{\mathrm{nc}}_A(D)$. So M^{nc}_A is not a semimeasure; it is the plausibility $\mathrm{Pl}_0(\psi_D)$ (Proposition 1.3), the dual of a belief function.*

(b) $M^{\mathrm{nc}}_A(D) \geq \varepsilon_\emptyset$ for every consistent D, so $-\log M^{\mathrm{nc}}_A \leq \log(1/\varepsilon_\emptyset)$ on all consistent data. The coin-free consistency-requiring weight is at least $2^{-|p_\emptyset|}$, and the Kolmogorov version $\min\{|p| : B \cup \mathsf{W}_{V(p)} \cup D^{\pm}$ consistent$\}$ is at most $|p_\emptyset|$ on all consistent D.

(c) The consistency-requiring posterior after D is the prior restricted to the tapes whose theory is consistent with D: data eliminate hypotheses but never change the odds between survivors.

Proof. (a) For consistent H, $H \cup \{\varphi^b\}$ is consistent iff $H \nvdash \varphi^{1-b}$, so both labels are possible iff H decides neither, and exactly one otherwise. (b) If $A_\omega = \emptyset$, then $H_\omega = B \cup D^{\pm}$ is consistent. (c) holds by definition. □

Part (c) is the one-tape form of Proposition 2.22(b) of Hänni and Claude (Anthropic) (2026). The inductive work is then done by adding the data to the hypothesis, and the proper predictor of this weight, Dempster's rule (Proposition 1.3), pays for it:

Proposition 1.20 (The consistency-requiring predictor)**.** *(a) For every consistent D and $\varphi \notin \operatorname{dom} D$ with $B \cup D^{\pm} \nvdash \varphi^b$: $\mathrm{Bel}^{\mathrm{nc}}_D(\varphi^b) \leq 1 - \varepsilon_\emptyset / M^{\mathrm{nc}}_A(D) \leq 1 - \varepsilon_\emptyset$. So on every consistent reveal sequence the consistency-requiring cumulative loss $\sum_t -\log \mathrm{Bel}^{\mathrm{nc}}_{D_{t-1}}(\varphi_t^{b_t})$ is at least $\log \frac{1}{1-\varepsilon_\emptyset}$ times the number of steps whose datum is not entailed by the earlier data.*

(b) In the literal setting, let a_i and $s_{i,j}$ $(i, j \geq 1)$ be fresh atoms, computable in (i, j), and put $e_{i,j} := \neg s_{i,1} \wedge \cdots \wedge \neg s_{i,j-1} \wedge s_{i,j}$ and $b_{i,j} := a_i \vee e_{i,j}$. (b1) For $D_j := \{(a_1, 1), (b_{1,j}, 1)\}$ and all but at most $1/(\mathrm{Bel}_0(a_1)\varepsilon_\emptyset)$ indices j, the consistency-requiring probability of D_j is strictly larger in the order $(a_1, b_{1,j})$ than in the order $(b_{1,j}, a_1)$. (b2) Let $c_R := |p_R|$ for a V-program p_R of an index of the deterministic machine emitting $\forall x\, R(x)$. For every n there are indices $j_1, \ldots, j_n \leq \lceil 2^{c_R+1}/\varepsilon_\emptyset \rceil$ such that, for $D_n := \{(a_i, 1), (b_{i,j_i}, 1) : i \leq n\}$, the consistency-requiring cumulative loss in the order $(b_{1,j_1}, a_1, b_{2,j_2}, a_2, \ldots)$ exceeds that in the order $(a_1, b_{1,j_1}, a_2, b_{2,j_2}, \ldots)$ by at least $n \log(1 + 2^{-c_R}\varepsilon_\emptyset/2)$ bits.

In (a) the empty theory lies in the denominator of $\mathrm{Bel}^{\mathrm{nc}}_D$ and not in its numerator. For fresh atoms α_j the stream $(\alpha_j, 1)_{j \geq 1}$ has no entailed datum, so the consistency-requiring loss on n data is at least $n \log \frac{1}{1-\varepsilon_\emptyset}$, while $\{\forall x\, R(x)\}$ decides the stream and the proof-requiring loss is $O(1)$ (Theorem 1.11). In (b) the probabilities of the two orders of one pair differ by a term whose sign the universal law does not fix for a single pair; the events in which the hypothesis proves $b_{i,j}$ without proving a_i are disjoint in j, so most j have the sign that favours revealing the stronger datum first (**??**). Under normalised proof-requiring prediction the opposite holds: revealing the weaker datum first is more probable (Theorem 1.26). The predictor is a mixture of the proof-requiring predictor and a component of weak theories together with the data (**??**).

The graded weight $M_A^{1/2}$ dominates M_A, but it is not a law of theories:

Proposition 1.21 (Graded consistency-requiring weight)**.** *(a) For consistent A, $\nu_A^{1/2}$ is a weak semimeasure with $\nu_A^{1/2}(\emptyset) = 1$, and is order independent. Hence $M_A^{1/2}$ is a weak semimeasure*

with $M_A^{1/2}(\emptyset) = 1 - \mathrm{Inc}_0$*, and the conditionals of* $M_A^{1/2}/M_A^{1/2}(\emptyset)$ *form an order-independent predictor.*

(b) $M_A^{1/2} \geq M_A$*.*

(c) It charges for entailed data: for fresh atoms p, q *and* $D := \{(p, 1)\}$*,* $M_A^{1/2}(D + (p \vee q, 1)) \leq M_A^{1/2}(D) - \varepsilon_\emptyset/4$*, although* $B \cup D^\pm \vdash p \vee q$*. So* $M_A^{1/2}$ *is not the law of any theory machine.*

(d) It depends on how the data are packaged: $M_A(\{(p \wedge q, 1)\}) = M_A(\{(p, 1), (q, 1)\})$*, but* $M_A^{1/2}(\{(p \wedge q, 1)\}) - M_A^{1/2}(\{(p, 1), (q, 1)\}) \geq \varepsilon_\emptyset/4$*.*

(e) $M_A^{1/2}/M_A^{1/2}(\emptyset)$ *is not a containment functional, not even of a random partial function on* Sent *that ignores logic. For every sentence* φ*, the Möbius sum on the two inputs* $\varphi, \neg\varphi$ *(the companion paper's Proposition A.2) equals* $-\frac{1}{2}\mathbb{U}\{\omega \in \mathrm{Cons} : A_\omega$ *decides neither* φ *nor* $\neg\varphi\}$ *for* $M_A^{1/2}$*, which is at most* $-\frac{1}{2}\varepsilon_\emptyset < 0$ *when* B *does not decide* φ*; for* M_A *the same sum equals* $\mathbb{U}\{\omega \in \mathrm{Cons} : A_\omega$ *decides neither*$\} \geq 0$*.*

Each theory contributes a product over the data of 1 for a proved datum and $\frac{1}{2}$ for a tolerated one; the empty theory supplies the $\varepsilon_\emptyset/4$ of (c) and (d), and a theory that decides neither φ nor $\neg\varphi$ contributes $1 - 2 + \frac{1}{2} = -\frac{1}{2}$ to the Möbius sum of (e) (**??**). The advantage of $M_A^{1/2}$ over M_A can come within $2|D| + O(\log)$ bits of $K(\mathrm{dom}\, D) + |D|$ (**??**), which measures packaging, not better induction. Finally, the distinction between the two kinds of compatibility exists only because the random theory is incomplete:

Proposition 1.22. *For a complete consistent theory* T *(that is,* $B \cup T$ *complete and consistent) and every data set* D*:* $B \cup T \vdash \psi_D$ *iff* $B \cup T \cup D^\pm$ *is consistent. Hence for any prior concentrated on complete theories, proof-requiring and consistency-requiring compatibility coincide, the trichotomy has no independent outcome, and conditioning on* D *restricts to the complete theories that contain* $D^\pm$*.*

Proof. A complete consistent $B \cup T$ proves ψ_D or $\neg\psi_D$, and is consistent with ψ_D iff it does not prove $\neg\psi_D$. □

In short: under proof-requiring compatibility a theory must explain every datum; under consistency-requiring compatibility it need only tolerate it, and toleration is too weak to discriminate between theories. Priors of the Demski type, which complete their random theory, cannot see the difference (**??**).

1.7 Limits and identification

Let D_∞ be a consistent stream, $E_\infty := \bigcap_n E_{D_n}$ and $P_n := \mathbb{U}(\cdot \mid E_{D_n})$. If $\mathbb{U}(E_\infty) > 0$, the posterior converges in total variation to $P_\infty := \mathbb{U}(\cdot \mid E_\infty)$, the prior restricted to the tapes whose theory is a consistent extension of the data, complete or not, and the loss is bounded; otherwise the loss diverges (**??**). What P_∞ is depends on whether the truth is complete:

Proposition 1.23 (Limits)**.** *Let* T^* *be consistent and r.e.*

(a) Complete decidable truths are identified up to equivalence. *If* $B \cup T^*$ *is complete (hence decidable) and* $\mathrm{Cn}(D_\infty^\pm) = \mathrm{Cn}(T^*)$*, then* $P_n\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\} \to 1$*,* $\sup_\chi |\mathrm{Bel}_{D_n}(\chi) - \mathbf{1}[T^* \vdash \chi]| \to 0$*,* $\sup_\chi \mathrm{Indep}_{D_n}(\chi) \to 0$*, and the cumulative loss converges to* $-\log(\mathbb{U}_A[T^*]/M_A(\emptyset)) \leq K_\equiv(T^*) + c$*. If* T^* *is axiomatised by one sentence* τ *and* $(\tau, 1) \in D_n$*, the first probability is* 1 *from then on.*

(b) Incomplete truths, two-valued data. *If $B \cup T^*$ is incomplete and $\mathrm{Cn}(D^{\pm}_{\infty}) = \mathrm{Cn}(T^*)$, then P_∞ is the prior restricted to the consistent extensions of T^*, and for every ψ independent of $B \cup T^*$: $\lim_n \mathrm{Bel}_{D_n}(\psi) = \mathbb{U}_A(\mathrm{Ext}(T^* + \psi))/\mathbb{U}_A(\mathrm{Ext}T^*) \in (0,1)$, $\lim_n \mathrm{Dis}_{D_n}(\psi) = \mathbb{U}_A(\mathrm{Ext}(T^* + \neg\psi))/\mathbb{U}_A(\mathrm{Ext}T^*) \in (0,1)$ and $\lim_n \mathrm{Indep}_{D_n}(\psi) \geq \mathbb{U}_A[T^*]/\mathbb{U}_A(\mathrm{Ext}T^*) > 0$. The most probable limit class can be a strict extension: in the literal setting, with fresh atoms α_i on a computable sequence of distinct words, $U := \{\alpha_i : i \in \mathbb{N}\}$ and $T_j := U \setminus \{\alpha_j\}$, for all but finitely many j the limit posterior on data presenting T_j gives the class of U more probability than that of T_j; a most probable class exists, and it strictly extends T_j.*

(c) Three-valued data identify. *If every sentence appears with its T^*-trichotomy label, then $\mathbb{U}(\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\} \mid E^3_{D_n}) \to 1$.*

A complete consistent theory has no proper consistent extension, which gives (a); in (b) the classes $[T_j]$ are pairwise distinct, so their prior masses sum to at most 1 and tend to 0, while U is simple (**??**). Part (b) refutes the natural expectation that the most probable limit class is the truth's. The failure to identify incomplete truths from two-valued data is not an artefact of the prior; it is the Bayesian shadow of Gold's distinction between learning from text and from an informant (cf. Gold, 1967):

Proposition 1.24 (Text against informant)**.** *In the literal setting, with fresh atoms α_i as above, let $\Lambda_k := \{\alpha_i : i < k\}$ and $\Lambda_\infty := \{\alpha_i : i \in \mathbb{N}\}$. Call a sequence (σ_t) of sentences a* text *for T if $\mathrm{Cn}(\{\sigma_t : t\}) = \mathrm{Cn}(T)$. No function F from finite sequences of sentences to deductive classes, computable or not, identifies in the limit, on every text, every member of $\{[\Lambda_k] : k \geq 1\} \cup \{[\Lambda_\infty]\}$.*

The proof is the locking argument of Gold (1967) (**??**). So no learner identifies all consistent r.e. theories from two-valued data, while three-valued data, an informant, identify every consistent r.e. T^* in the limit posterior (Proposition 1.23(c)), non-computably. Above arithmetic the random theory is almost never complete:

Proposition 1.25 (Random theories above Q are almost surely incomplete)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$.*

(a) For every theory machine G: $\mathbb{U}\{\omega : B \cup A^G_\omega$ is consistent, proves every axiom of Q and decides every $\mathcal{L}_{\mathrm{ar}}$-sent 0. In particular, if $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, the consistent random theory of M_A is almost surely incomplete, and every posterior gives probability 0 to complete theories.

(b) For every function machine F with inputs $\mathrm{Sent}_{\mathcal{L}_{\mathrm{ar}}}$ and labels $\{0,1\}$ (the companion paper's Definition 3.6): $\mathbb{U}\{\omega : f^F_\omega$ is total, labels every Q-provable sentence 1 and every Q-refutable sentence $0\} = 0$; likewise with PA, or any consistent r.e. extension of Q, in place of Q.

On the event of (a), "the first of $\sigma, \neg\sigma$ found provable" is a total separator of the theorems of Q from its refutable sentences; a majority vote over a positive-measure set of tapes (**??**) would make it computable, which the effective inseparability of that pair excludes (standard; see Smullyan, 1961). This is a new application of the fact that separating sets of effectively inseparable pairs have measure zero. Part (b) says that random labellers almost surely fail consistent guessing. A consequence is that every theory-machine law above Q has a positive local defect somewhere (**??**).

1.8 Order dependence and the Kolmogorov predictor

The semimeasure-convention loss of M_A is order independent (Proposition 1.1(d)). The normalised loss is not, already on two data:

Theorem 1.26 (Order dependence in the normalised convention)**.** *(a) Let a, c be sentences with $B \cup \{a\}$ consistent and $B \cup \{\neg a\}$ consistent and deciding neither c nor $\neg c$, and let $D := \{(a,1),(a \vee c,1)\}$. The normalised probability of D is strictly larger in the order $(a \vee c, a)$ than in the order $(a, a \vee c)$ (with the explicit ratio (**??**) of **??**), while the loss in the semimeasure convention is the same in both orders.*

(b) Let a_i, c_i ($i \geq 1$) be fresh atoms. For every n there are $y \in \{0,1\}^n$ and two reveal orders of the $2n$ data $\{(a_i^{y_i},1),(a_i^{y_i} \vee c_i,1) : i \leq n\}$ whose normalised losses differ by at least $2^{-c_1-1}n$ bits, where c_1 is the length of a V-program for the machine that emits fair-coin facts on the a_i.

Whether the random theory decides $a \vee c$ depends on its answer to a: a theory proving a always decides $a \vee c$, one refuting a need not. The witnesses use entailment between data, so the theorem does not answer the companion paper's question on logically independent data (**??**). The single shortest compatible theory can be a poor predictor even where the mixture is good; the bound of Proposition 1.13(b) is essentially attained:

Theorem 1.27 (The Kolmogorov predictor against Bayes)**.** *Let $a_1, \ldots, a_m$ be fresh atoms, $\mathrm{desc}(w) := \bigwedge_{i \leq m} a_i^{w_i}$ for $w \in \{0,1\}^m$, and fix any well-order $\prec$ of $\operatorname{dom} V$. There are a data set D of $2^m - 1$ sentences, all of the form $\neg\mathrm{desc}(v)$ labelled true, and a reveal order of D, such that (i) the Kolmogorov predictor with $\prec$ errs or abstains at every one of the $2^m - 1$ steps; (ii) $K_A(D) \leq m + K(m) + c$, and the Bayes predictor of Proposition 1.13(a) makes at most $K_A(D) + c \leq m + K(m) + c'$ mistakes on the same sequence. If $\prec$ refines length, then $m \leq K_A(D) \leq m + K(m) + c$, so with $k := K_A(D)$ the number of errors or abstentions is at least $2^{k-K(m)-c} - 1 \geq 2^{k-2\log k-c'} - 1$.*

An adversary presents $\neg\mathrm{desc}(v)$ for a valuation v left open by the current shortest compatible theory, keeping the valuation whose shortest theory comes last (**??**). The stream is not computable. As in the companion paper's conjunction ladder (its Theorem 9.10), it is the selection rule that fails, not the prior; in SOI on strings the decoys would be beaten by the constant labeller "true", which is unavailable here because $\{\neg\mathrm{desc}(v) : v \in \{0,1\}^m\}$ is inconsistent.

We close with the main open question about the object itself. Call a Borel random variable Θ whose values are consistent, deductively closed sets of sentences a *random consistent theory*, and $\beta(\chi) := \Pr[\chi \in \Theta]$ its *provability function*.

Question 1.28. Does M_A dominate every lower semicomputable provability function, that is, is $\mathrm{Bel}_0 \geq 2^{-c_\beta}\beta$ whenever β is lsc?

Domination holds with the slack $K(S)$ of Proposition 1.17, and for the closure of a theory machine's output; on random sets of literals of a fresh predicate the question is equivalent to the companion paper's Question 3.14 for binary labels (**??**). The obstacle is the companion paper's: a lsc law of a random object need not be realisable by a machine that reads one tape. Note that Bel_0 itself is not lsc (Theorem 1.5).

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